



# Biofilm Formation and Associated Gene Expression in Multidrug-Resistant *Klebsiella pneumoniae* Isolated from Clinical Specimens

Priyanka Ashwath<sup>1</sup> · Vijaya Kumar Deekshit<sup>1</sup> · Anusha Rohit<sup>2,3</sup> · I. Dhinakaran<sup>2</sup> · Iddya Karunasagar<sup>3</sup> · Indrani Karunasagar<sup>1,3</sup> · Dharnappa Sannejal Akhila<sup>1</sup>

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## Abstract

Biofilms reduce the bacterial growth rate, inhibit antibiotic penetration, lead to the development of persister cells and facilitate genetic exchange. The biofilm-associated *Klebsiella pneumoniae* infections have not been well studied, and their implications in overcoming the effects of antimicrobial therapy are yet to be fully understood. Hence this study evaluated the antibiotic resistance pattern, antibiotic resistance determinants of extended-spectrum beta-lactamase (ESBL) family. Biofilm-forming ability of seventy multidrug-resistant clinical isolates of *K. pneumoniae* and the biofilm-associated genes of representative biofilm formers from a tertiary care hospital were also assessed. The *K. pneumoniae* isolated from urine exhibited resistance towards ceftazidime, nalidixic acid and meropenem. Isolates from blood were resistant to cefuroxime. Higher rates of resistance were observed towards cefuroxime, nalidixic acid, and meropenem for the isolates from the endotracheal aspirate. Extended spectrum beta-lactamase production by CLSI's disc diffusion-based confirmation test revealed all the *K. pneumoniae* to be as ESBL producers. Most of the isolates harboured the *bla* gene variants, *bla*<sub>SHV</sub> and *bla*<sub>TEM</sub>. Majority of the isolates were colistin sensitive. 97.1% of the *K. pneumoniae* produced biofilm. *K. pneumoniae* isolated from pus and blood produced fully established biofilms. Strong biofilm formers were sensitive to co-trimoxazole and ciprofloxacin. Moderate biofilm formers exhibited sensitivity towards meropenem and imipenem. Expression of the *fimH* gene was increased, while *mrkD* showed reduced expression among the strong biofilm formers. Moderate biofilm formers showed variable expression of the genes associated with the biofilm formation. The weak and non-biofilm formers showed reduced expression of both the fimbrial genes. Multidrug-resistant isolates produced ESBLs and formed well-established biofilms.

## Introduction

*Klebsiella pneumoniae* is the most significant pathogenic gram-negative, nonmotile bacillus causing urinary tract infections, pneumonia, bacteraemia, pyogenic liver abscesses and wound infections [1]. The *K. pneumoniae* infections are a serious global health concern,

threatening the hospitalized and immune-compromised patients [2]. Previous studies have revealed that *K. pneumoniae* is involved in 4 to 15% cases of septicaemia, 2 to 4% of wound infections and 3 to 30% neonatal septicaemia [3]. This pathogen is recognized for the frequency and diversity of antimicrobial resistance (AMR) genes [4]. Pathogenicity of *K. pneumoniae* has been reported to be due to the presence of several other virulence factors such as capsular polysaccharides, lipopolysaccharide (LPS), iron-scavenging systems (siderophores), hypermucoviscosity, capsule synthesis and adhesion molecules [5]. The nosocomial infections caused by *K. pneumoniae* are facilitated by the ability of the organism to form biofilm which protects the pathogen from the host immune responses, antibiotics, and enhances its persistence on host epithelial tissues and medical device surfaces [6]. Biofilm is an aggregate of microorganisms attached to an inert or living surface by a self-produced exo-polymeric matrix

✉ Dharnappa Sannejal Akhila  
akhila@nitte.edu.in

<sup>1</sup> Nitte (Deemed to be University), Nitte University  
Centre for Science Education & Research, Deralakatte,  
Mangalore 575018, India

<sup>2</sup> Department of Microbiology, Madras Medical Mission,  
Chennai, India

<sup>3</sup> Nitte (Deemed to be University), University Enclave,  
Medical Sciences Complex, Mangalore, India

which includes proteins, polysaccharides and extracellular DNA (exDNA). Biofilms reduce the bacterial growth rate, inhibit antibiotic penetration, lead to the development of persister cells and facilitate genetic exchange [7, 8]. The mechanism of biofilm formation among the clinical strains of *K. pneumoniae* is reported to be associated with a plethora of genes. Few significant genes include aerobactin (*iutA*), allantoin (*allS*), type I fimbriae (*fimA* and *fimH*), type III fimbriae (*mrkA* and *mrkD*), capsular polysaccharide (CPS) (*cpsD*, *treC*, *wabG*, *wcaG*, *wzc*, *k2A*, and *wzyK2*), quorum sensing (QS) (*luxS*), and colonic acid (*wcaJ*) [9–11]. In *Klebsiella* and *E. coli*, the type 1 fimbriae play a role in establishing urinary tract infections [12]. Type 3 fimbriae can efficiently mediate the binding of *Klebsiella* to the epithelial and endothelial cells of the respiratory and urinary tracts. They were reported to promote adherence and biofilm formation on inert surfaces [13]. The biofilm-associated *K. pneumoniae* infections have not been well studied and their implications in overcoming the effects of antimicrobial therapy are yet to be fully understood [14]. *Klebsiella* is recognized as a reservoir of antibiotic resistance genes that can spread to other Gram-negative bacteria. Furthermore, many genes responsible for drug resistance have been commonly found in the clinical isolates of *Klebsiella* [15]. Clinical manifestations of *K. pneumoniae* are pronounced further due to the emergence of drug-resistant strains characterized by the presence of carbapenemases and extended-spectrum- $\beta$ -lactamases (ESBL). ESBLs are rapidly evolving group of  $\beta$ -lactamases that catalyse the hydrolysis of third-generation cephalosporins by the acquisition of ESBL genes and produce related enzymes posing a major threat to the existing therapeutics [16]. There are greater chances of failure in the treatment of antibiotics against active ESBL-producing *K. pneumoniae*. Several studies have addressed the correlation between the ESBL production and pathogenicity in *Klebsiella* [17]. Horizontal transfer of antibiotic resistance genes via mobile genetic elements, such as plasmids and transposons help in the spread of ESBL [18]. *K. pneumoniae* isolates resistant to quinolones, aminoglycosides and third-generation cephalosporins have been reported causes for hospital outbreaks, other mechanisms of resistance facilitate the survival of the bacteria in nosocomial environments [19]. Therapeutic options for infections due to multidrug-resistant *K. pneumoniae* are not well established. Thus, a better understanding of the correlation between the resistance profile, ESBL production, and biofilm formation of *K. pneumoniae* is mandatory. The present study aimed to evaluate the resistance pattern of *K. pneumoniae* isolated from various clinical sources and to determine their ability to produce ESBL. In addition, the biofilm-forming ability and the associated gene expression among the biofilm formers were evaluated.

## Materials and Methods

### Screening for *Klebsiella pneumoniae* Isolates and Antibiotic Susceptibility Testing

A total of seventy *K. pneumoniae* from endotracheal aspirate ( $n = 10$ ), urine ( $n = 33$ ), blood ( $n = 14$ ), pus ( $n = 8$ ), wound swabs ( $n = 3$ ), and tissue ( $n = 2$ ) isolated at the Madras Medical Mission, Chennai were used in the study (Suppl. Table 1). The *K. pneumoniae* isolates were identified and antibiotic susceptibility testing was performed using Vitek2 compact system (bioMerieux). The morphologically confirmed *K. pneumoniae* isolates were stored in Luria Bertani (LB) broth (HiMedia, India) containing 30% glycerol at  $-80^{\circ}\text{C}$ .

### Determination of ESBL Production

The phenotypic confirmation of ESBL production among the potential ESBL-producing isolates was performed by double-disc diffusion test using ceftazidime (30  $\mu\text{g}$ ) alone and in combination with clavulanic acid (10  $\mu\text{g}$ ) placed 25 mm apart. Isolates showing an increase in zone diameter of  $\geq 5$  mm around the clavulanate combined discs compared to that of the disc alone were considered ESBL producers.

### Molecular Characterization

#### Genomic DNA Extraction and Polymerase Chain Reaction

Genomic DNA of all the *Klebsiella* isolates was extracted using the cetyl trimethyl ammonium bromide (CTAB) method as described by Ausubel et al. [20]. PCR was performed in a 30  $\mu\text{l}$  mixture containing 3  $\mu\text{l}$  of 10 X buffer (100 mM of Tris-HCl with a pH 9, 1.5 mM of  $\text{MgCl}_2$ , 500 mM of KCl, and 0.1% gelatin), 200  $\mu\text{M}$  of deoxyribonucleotide triphosphate (dATP, dTTP, dGTP and dCTP), 10 pM each of forward and reverse primers, and 1.0 unit of *Taq* DNA polymerase. All the isolates were tested for the identification of genus *Klebsiella* targeting the *gyrA* gene. Species-specific confirmation of *K. pneumoniae* was done by targeting the 16S rDNA-23S rDNA internal transcribed spacer (ITS) region (Primers details are provided in Suppl table 2). Further, the 16S rRNA-23S rDNA ITS region of the representative *K. pneumoniae* samples included in the study were sequenced. The obtained sequences were compared with the *K. pneumoniae* reference strains from the

GenBank repository. Results of the type strain accession number, similarity to type strain, and sequence length are provided in (Suppl table 3).

### Detection of Antibiotic Resistance Genes

Presence of antibiotic resistance genes coding for ESBL (*bla*<sub>CTX-M</sub>, *bla*<sub>SHV</sub>, *bla*<sub>TEM</sub>). Plasmid-mediated quinolone resistance (*qnrA*, *qnrB*, *qnrS*, *qepA*) and chloramphenicol resistance (*catA1*) were investigated by PCR (Primer details are provided in Suppl table 4).

### Biofilm Formation Assay

The biofilm-forming ability of all the seventy isolates was evaluated using a 96-well microtiter plate method. Briefly, 10 µl of the overnight grown *K. pneumoniae* cultures were added to the wells containing 190 µl of LB broth and incubated at 37 °C for 24 h to allow the biofilm development. The contents of the wells were removed and washed with sterile distilled water. The cells were fixed with methanol, followed by staining with crystal violet (1%), and incubated for 15 min. Crystal violet bound to biofilm cells were solubilized by using glacial acetic (30%) acid, and absorbance was measured at 630 nm. The experiment was performed in triplicates. True OD values were obtained by subtracting the control value. Classification of isolates was done as follows  $0 \leq \text{O.D.c} = \text{non-biofilm former}$ ,  $0. \text{D.c} < 0. \text{D} \leq (2 * \text{O.D.c}) = \text{weak biofilm former}$ ,  $(2 * \text{O.D.c}) < 0. \text{D} \leq (4 * \text{O.D.c}) = \text{moderate biofilm former}$ ,  $4 * \text{O.D.c} < 0. \text{D} = \text{strong biofilm former}$  [21].

### Evaluation of Expression of the Genes Associated with Biofilm Formation

#### RNA Extraction by TRIzol Method

The representative isolates were grown in triplicates in cell culture plate (6 well) with LB medium (negative control), the biofilm formers and *K. pneumoniae* ATCC 10031 (positive control) for 24 h at 37 °C. RNA was extracted from two representative isolates of the biofilm-forming category: Strong (B5186, P4399), moderate (F15041, 03386), weak (U3067, BAN510) and non-biofilm formers (P6324, U5041). 1.5 ml of the culture were lysed by the addition of 1 ml Trizol reagent. After incubation, 200 µl of chloroform was added and centrifuged at 12,000×g at 4 °C for 15 min. RNA was precipitated from the aqueous phase by using 0.5 ml of isopropyl alcohol. The obtained pellet was washed with 75% ethanol and resuspended in 30 µl of nuclease-free water. The concentration and purity were analysed by using a spectrophotometer (Biospectrometer, Eppendorf, Germany).

### DNase Treatment and cDNA Synthesis

DNase treatment was performed using a Thermo scientific kit. RNA sample (0.5 µg) was mixed with 1 µl 10× reaction buffer with MgCl<sub>2</sub>, 1 µl of DNase. Volume was made up to 10 µl using DEPC-treated water. Samples were incubated at 37 °C for 30 min followed by deactivation at 65 °C for 10 min. Purity of the obtained RNA was measured using spectrophotometer and stored at −80 °C for further use. The RNA samples were normalized to 500 nanograms for cDNA synthesis. Further, reverse transcription of the RNA samples was performed by Prime Script one-step reverse cDNA synthesis kit (TaKaRa, Japan) with slight modifications. cDNA samples were confirmed by PCR for the expression of biofilm-forming genes and stored at −20 °C.

### Quantification of Biofilm-Forming Genes by Real-Time PCR (q-PCR)

Relative gene expression of the three genes (*fimH*, *mrkA*, and *mrkD*) reported to be involved in the biofilm formation of *K. pneumoniae* was analysed. The gene sequences for biofilm-forming genes were retrieved from NCBI GenBank. Primers were designed by using the software PRIMER 3 (Suppl table 5). *rpoA* was used as a housekeeping gene to standardize the relative expression of the target gene. Appropriate concentrations of primers for all the genes were determined following standardization, and optimized concentrations were used in the subsequent step. SsoAdvanced universal SYBR green supermix (BioRad, CA, USA) was used for the q-PCR analysis. The real-time PCR master mix (10 µl) contained 5 µl advance 2× universal SYBR green supermix (BioRad, CA, USA), 200 nM of forward and reverse primers, 1 µl of template RNA, and the final volume was adjusted with molecular grade water. The reaction was performed in real-time PCR (CFX96 Touch™ Real-time PCR detection system, BioRad, USA) The cycling conditions were as follows: initial denaturation of 95 °C for 3 min, followed by 45 cycles of denaturation at 95 °C for 10 s, 55 °C for 20 s, and extension of 72 °C for 20 s. Analysis of the melt curve was performed to check the amplification of untargeted regions. Relative quantification of all the genes was calculated using the  $2^{-\Delta\Delta\text{ct}}$  formula by Livak and Schmittgen [22].

### Data Analysis

For the real-time PCR analysis, an unpaired student *t*-test was performed. All the calculations were performed using Microsoft Excel 2019 and any value < 0.05 was considered significant.

## Results

### *K. pneumoniae* Isolates and Antibiotic Susceptibility Testing

During the study period, a total of seventy non-repetitive clinical of *Klebsiella* isolates from various clinical specimens included endotracheal aspirate, urine, blood, pus, tissue, and wound swabs. All the 70 isolates that were *Klebsiella pneumoniae* were identified and confirmed by Vitek 2 compact (bioMérieux, France) system.

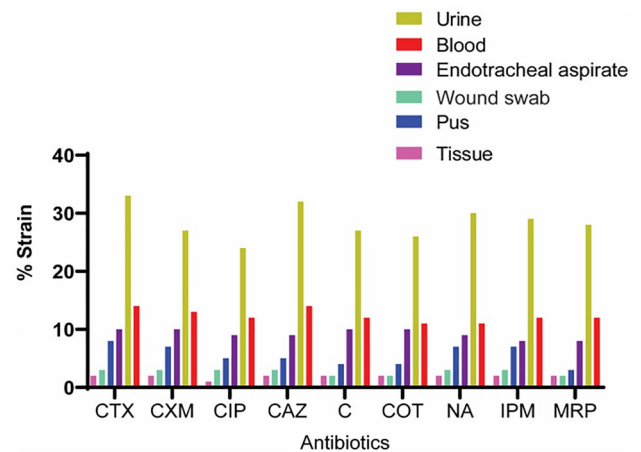
### Molecular Characterization of the *K. pneumoniae* Isolates

Species confirmation done by targeting the 16S rDNA -23S rDNA internal transcribed spacer (ITS) region confirmed all the seventy isolates to be as *K. pneumoniae*. Comparison of the 16S rRNA-23S rDNA ITS region sequences in the study with the type strains further confirmed the isolates to be *K. pneumoniae*. NCBI BLAST revealed that the sequences from the representative isolates included in the study shared 100% similarity with the type strains deposited in the GenBank repository (Suppl table 3). However, the sequences included in the study did not show any similarity to the other members of *Enterobacteriaceae*, as well as other species of the genus *Klebsiella*.

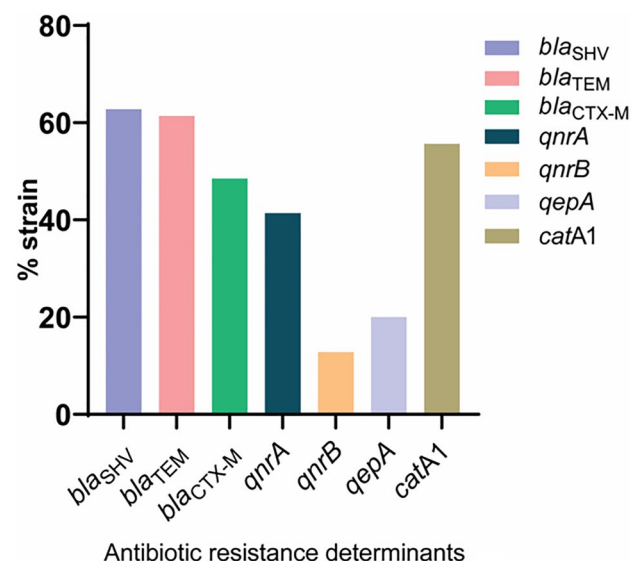
Antibiotic susceptibility testing for all the isolates demonstrated, 100% resistance to cefotaxime followed by cefuroxime 68 (96.8%), nalidixic acid 67 (96%), ceftazidime 65 (95.3%), imipenem (90%), meropenem 62 (89%), ciprofloxacin 62 (88%), chloramphenicol 59 (84%) and co-trimoxazole 58 (82%). Majority of the isolates were from urine, blood, followed by the endotracheal aspirate. *K. pneumoniae* isolated from urine exhibited resistance towards ceftazidime, nalidixic acid, and meropenem likewise isolates collected from blood were resistant to cefuroxime. Those isolates from endotracheal aspirate exhibited higher resistance towards cefuroxime, nalidixic acid, and meropenem. Source of isolation and the antibiogram of the isolates have been represented in Fig. 1. Colistin susceptibility was then tested by Vitek II Compact (bioMérieux, France) and most of the isolates tested were sensitive as per CLSI standards. All the seventy isolates tested for the ESBL production were ESBL producers.

### Detection of Antibiotic Resistance Genes

ESBL encoding genes were detected including *bla<sub>SHV</sub>* 45 (63%), *bla<sub>TEM</sub>* 43 (61%), and *bla<sub>CTX-M</sub>* 34 (49%). Among the plasmid-mediated quinolone resistance genes, 41.4%



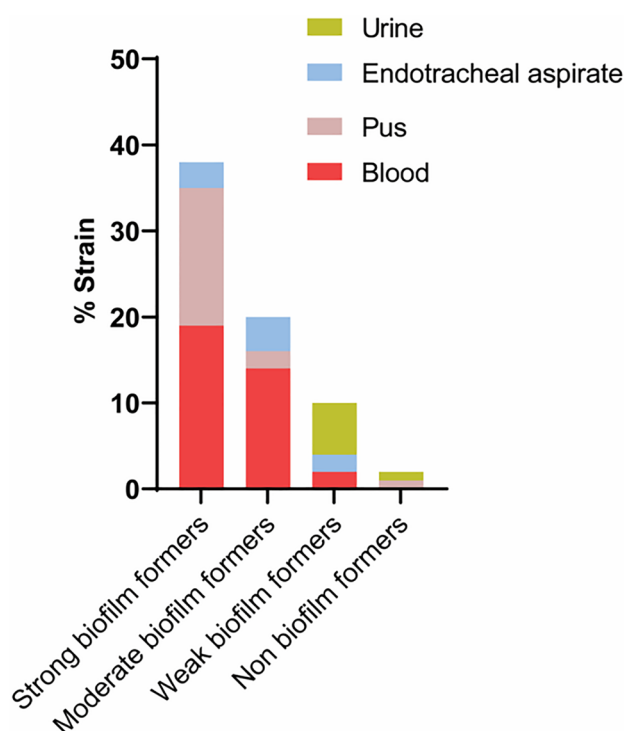
**Fig. 1** Source-wise representation of antibiotic resistance pattern of all the clinical isolates of *K. pneumoniae* against nine antibiotics. CIP ciprofloxacin, CAZ ceftazidime, CXM cefuroxime, C chloramphenicol, COT co-trimoxazole, CTX cefotaxime, IPM imipenem, MRP meropenem, NA nalidixic acid



**Fig. 2** Graphical representation of various antibiotic resistance determinants harboured by the clinical isolates of *K. pneumoniae* in the study

and 12.8% of the isolates were found to harbour *qnrA* and *qnrB* genes. None of the isolates carried the *qnrS* gene. 14 (20%) and 39 (55.7%) of the isolates possessed the *qepA* and *catA1* genes, respectively. Antimicrobial resistance determinants detected have been represented in Fig. 2.

Among the seventy isolates tested for their biofilm-forming ability, 68 (97%) of the isolates were biofilm producers. Of which, 38 (54%), 20 (29%) and 10 (14%) were strong, moderate and weak biofilm formers. 2 (2.8%) of the isolates were non-biofilm formers (Fig. 3).



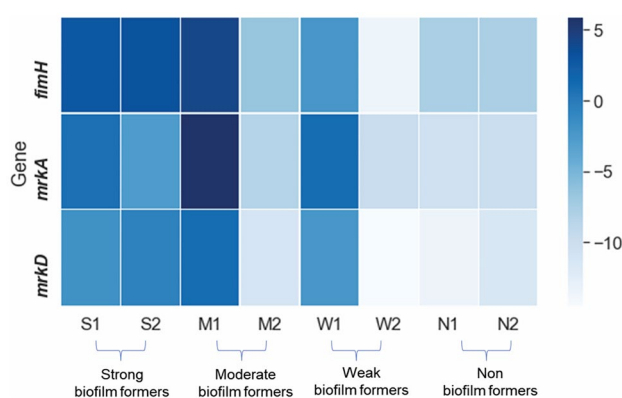
**Fig. 3** Graphical representation of the source and the biofilm-forming ability of the clinical isolates of *K. pneumoniae*

### Correlation Between the Biofilm-Forming Ability, Source of Isolation and Antibiotic Susceptibility Pattern of *K. pneumoniae*

The source of isolation of the majority of the strong biofilm formers was urine, pus and blood, whereas moderate biofilm formers were from blood. Source of isolation and biofilm-forming ability of *K. pneumoniae* have been represented in Fig. 3. The non-biofilm formers were resistant to all the drugs tested. 19 (50%) of the strong biofilm formers were resistant to all the antibiotics tested, while 11 (29%) and 8 (21%) of the strong biofilm formers were sensitive only towards co-trimoxazole and ciprofloxacin, respectively. 8 (40%) and 6 (30%) of the moderate biofilm formers were found sensitive only to meropenem and imipenem, respectively (Suppl. table 6), while 6 (30%) of the moderate biofilm formers were resistant to all the drugs included in the study. It was noteworthy that 80% of the weak biofilm formers were resistant to all the drugs, while 20% of the weak biofilm formers were sensitive to chloramphenicol.

### Evaluation of Gene Expression of the Biofilm-Forming Genes

Relative gene expression of the three biofilm-associated genes (*fimH*, *mrkA*, and *mrkD*) of *K. pneumoniae* was



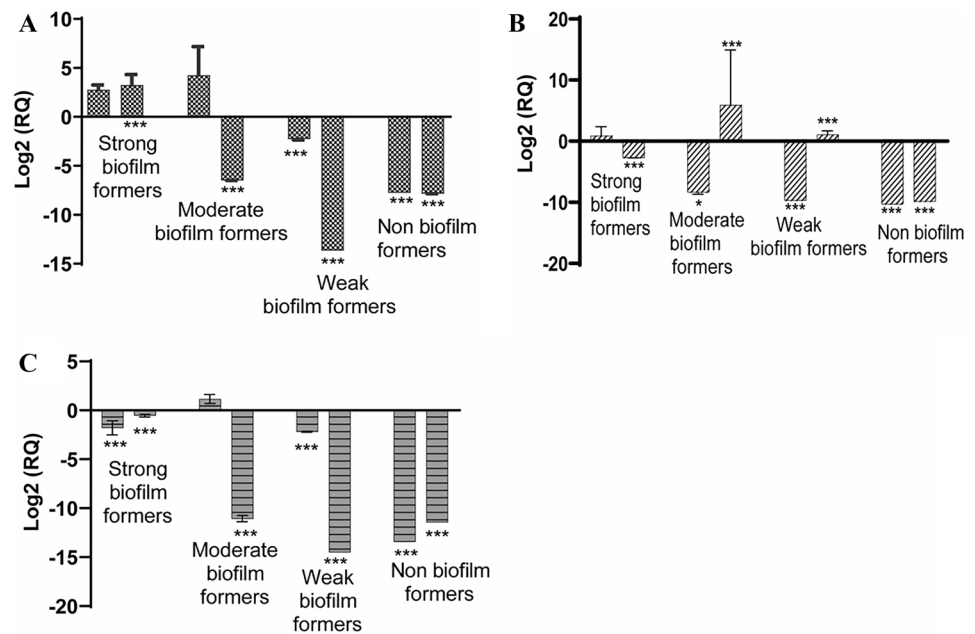
**Fig. 4** Differential expression of biofilm genes in the *K. pneumoniae* clinical isolates. The expression level of biofilm genes in the clinical isolates of *K. pneumoniae* is presented using fold-change values transformed to log2 format compared to the control. The log2 (fold-change values) and the colour scale are shown right to the heat map

analysed (Figs. 4, 5). The results were interestingly different among the strong, moderate, and weak biofilm formers concerning the type 1 (*fimH*) and type 3 (*mrkA* and *mrkD*) fimbrial adhesins. Strong biofilm formers (B5186, P4399) showed increased expression of the *fimH* gene, which mediates the binding of the bacteria to the surface. *mrkA* aids rapid biofilm formation and the gene *mrkD* facilitate dense biofilm formation in *K. pneumoniae*. The strong and moderate biofilm formers showed variable expression of the gene *mrkA*. Expression of the *mrkD* gene was seen to be reduced in the strong biofilm formers. Weak (U3067, BAN510) and the non-biofilm formers (P6324, U5041) exhibited reduction in the expression of type 1 and type 3 fimbrial genes.

### Discussion

In the current study, all the *Klebsiella* isolates obtained from various clinical sources were *K. pneumoniae*. Other studies also show a higher prevalence of *K. pneumoniae* among the clinical isolates compared to other species of *Klebsiella* [23, 24]. Carbapenems are believed to be the drug of last resort for the treatment of infections caused by multidrug-resistant pathogens. Gupta et al. [25] suggested imipenem as the most effective drug against *K. pneumoniae* infections. In the present study, 89.8% of isolates showed resistance to imipenem, of which 44% of the isolates were from urine, this is in line with the study conducted by Papadimitriou-Olivgeris et al. [26], Varghese et al. [27], and few reports [28, 29] indicated a similar trend. The third-generation cephalosporins (3GC) assumed to be the clinically relevant class of antibiotics in the treatment of serious infections caused by *K. pneumoniae* were also tested in this study. All the isolates were

**Fig. 5** Differential expression of biofilm genes in the *K. pneumoniae* clinical isolates. Expression of genes **A** *fimH*, **B** *mrkA*, and **C** *mrkD* in the strong, moderate, weak, and non-biofilm formers. The two bars in each of the biofilm-forming category indicate the representative isolates used for the expression studies (\*indicates  $P < 0.05$ , \*\*\* indicates  $P < 0.02$ ). The *rpoA* targeting the 16S rDNA was used as a reference gene, with the control level expression of 1 at 0



resistant to cefotaxime. 95.3% and 96.8% of the isolates were resistant to ceftazidime and cefuroxime, respectively. Fluoroquinolones were considered one of the effective antibiotics against *Klebsiella* infections a decade ago. In the recent past higher rates of resistance towards ciprofloxacin have been recorded. The data from the current study support this with 88% of resistance to ciprofloxacin, which is in agreement with other reports [30–33]. The menace of drug resistance could be exacerbated by self-medication, lack of awareness, and misuse of antibiotics, but could be controlled by knowledge on the resistance pattern of the organism.

In our study, the most prevalent ESBL encoding gene was *bla<sub>SHV</sub>* (63%) followed by *bla<sub>TEM</sub>* (61%), and *bla<sub>CTX-M</sub>* (49%). Feizabadi et al. [34] demonstrated a low prevalence of *bla<sub>SHV</sub>* and a higher prevalence of *bla<sub>TEM</sub>* and *bla<sub>CTX-M</sub>*, respectively. The finding of Sahoo et al. [35] indicates *bla<sub>TEM</sub>* as the most predominant gene followed by *bla<sub>SHV</sub>* *K. pneumoniae*.

The prevalence of fluoroquinolone resistance has strikingly increased in recent years, and reports note a remarkable increase among Gram-negative pathogens. Resistance to quinolones could be due to different mechanisms, but in this study, 41.4% and 12.8% of the isolates harboured *qnrA* and *qnrB* and none of them carried *qnrS*. These are generally plasmid borne genes that can spread rapidly. This finding is contradictory with the result of Badawy et al. [36] showing higher prevalence of *qnrB* and *qnrS*, but the absence of *qnrA*. Shams et al. [37] reported higher levels of *qnrB* and *qnrS*. Wang et al. [38] detected a low level of *qnrB* and *qnrS*. Thus, there seems to be wide

variation in the type of *qnr* genes in clinical isolates of *K. pneumoniae*.

The key step in the establishment of infection by nosocomial pathogens such as *K. pneumoniae* is their ability to form biofilms on the host surface. The formation of biofilm affects the activity of the antimicrobial agents making them less effective. In our study highest number of *K. pneumoniae* (97.1%) isolates formed biofilms. Cepas et al. [39] demonstrated 37.6% of the *K. pneumoniae* isolates to be biofilm producers. Present study recorded majority of *K. pneumoniae* isolates to be strong and moderate biofilm formers. This is in contrast with few reports, where clinical isolates of *K. pneumoniae* exhibiting strong biofilm formation were less, in comparison to the moderate and weak biofilm formers [40–42]. The reason for the difference may be the sample size, source of isolation, and geographical region.

When considered the source of isolation and biofilm formation, we noted majority of the biofilm producers were from pus, blood and urine. Yang and Zhang, [43] noted that 62.5% of the biofilm *K. pneumoniae* were from blood, urine, sputum and wound swab. The biofilm-forming ability may not be necessarily associated with the source of isolation. As per our observation, regardless of the isolation source, any strain might be strong, moderate and weak biofilm formers.

The biofilm-forming ability demonstrates increased resistance towards antibiotics. In this study 40% and 30% of the moderate biofilm formers exhibited sensitivity only towards meropenem and imipenem, respectively. This is in contrast to the finding of Mishra et al. [44] who reported Gram-negative bacteria resistant to meropenem exhibited higher capacities to form biofilm. 50% of the strong biofilm

formers in the study were resistant to all the drugs. We noted that 21% of the strong biofilm formers were susceptible only to ciprofloxacin. This is in agreement with some of the reports [40, 45]. Only 29% of the strong biofilm formers exhibited sensitivity towards co-trimoxazole. As per the study by Subramanian et al. [46], the strong biofilm formers were resistant to co-trimoxazole. Possible reason for increased biofilm-forming ability of few isolates and decreased resistance towards certain drugs could be due to the mutations within the biofilm-forming genes, which could result in sensitivity of biofilm-forming strains without affecting their ability to form biofilms [47, 48]. All the non-biofilm formers were resistant to all the drugs tested in this study. This is contradiction to the study of Saha et al. [49] where all the biofilm-forming isolates were resistant in comparison to the non-biofilm formers.

According to Vutto et al. [50] plasmids harbouring the resistance genes would express the genes associated with adherence, biofilm formation or modulate biofilm-forming ability. Majority of the strong and moderate biofilm formers in our study were MDR strains. Thus, we possibly assume that the plasmids harbouring the resistance genes would aid them to modulate and form biofilm.

Type 1 fimbriae are one of the best described fimbrial adhesins in many bacteria of the family *Enterobacteriaceae*. Type 3 fimbriae, first identified and characterized in *K. pneumoniae*, is said to be necessary for biofilm formation on abiotic surfaces and also surfaces coated with host-derived materials including CAUTIs [51]. In our study *fimH*, *mrkA* and *mrkD* were detected in biofilm-forming *K. pneumoniae* from blood, pus, urine and endotracheal aspirate.

It is proposed that the expression of the type 1 fimbriae is regulated by the *fim*-switch regulatory mechanism containing the promoter for the major fimbrial subunit *fimA* [50, 52]. The nullification of type 3 fimbrial expression is compensated by the upregulation of the type 1 fimbriae and vice versa, which strongly indicates cross-regulation of the two fimbrial gene clusters. Supportive to this, findings by Stahlut et al. [13] suggest that type 1 and type 3 fimbriae both promote biofilm formation and also are able to compensate for each other. In our study, *fimH* showed increased expression, while there was reduced expression of the genes *mrkA* and *mrkD*.

Mutational studies in *K. pneumoniae* have showed type 3 fimbriae seem to promote biofilms in them [53]. In *E. coli*, the type 1 fimbriae have shown to promote biofilm formation [54]. But in the present study, reduced expression of the fimbrial genes did not have any impact on the formation of biofilms.

Few studies demonstrate that the strains unable to produce type 1 fimbriae efficiently can form biofilm, indicating the minimal role of fimbrial gene type 1 [53, 55], while some identified the equal contribution of type 1 and

3 fimbriae in biofilm formation [13, 52]. In the present study, the expression of type 1 fimbriae was enhanced, while expression of the type 3 fimbriae varied among the biofilm formers. However, different studies have come to contradictory conclusions with respect to the biofilm-forming genes. Hence, further studies are required to broaden the understanding of the co-relation between the biofilm formation and associated genes.

## Conclusion

*K. pneumoniae* has reached high levels of endemicity in hospitals and is one of the main causes of nosocomial outbreaks. In our study we report high prevalence of *K. pneumoniae* resistant to multiple antibiotics and were potent ESBL producers. Antimicrobial resistance can enhance biofilm-forming ability in several bacterial species. In this study biofilm-forming *K. pneumoniae* isolates demonstrated resistance to a wide range of antibiotics. The biofilm producers in the study were found to harbour both type 1 and type 3 fimbrial genes and were from urine, blood and pus. The findings from the study suggest the possible association between drug resistance and biofilm formation in the clinical isolates of *K. pneumoniae*. This could be considered as a major challenge for the treatment of *K. pneumoniae*-associated infections. Future studies could elaborate new concepts in preventing nosocomial *K. pneumoniae* infections.

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s00284-022-02766-z>.

**Author Contributions** PA—AM and BB. DVK—AM and GR. AR—AM and GR. ID—AM. IK—BB. IK—BB. ADS—AM and DB.

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**Data Availability (Data Transparency)** Not applicable.

**Code Availability (Software Application or Custom Code)** Not applicable.

## Declarations

**Conflict of interest** The authors declare that they have no conflict of interest.

**Ethical Approval** This study was approved by the Institutional Ethics Committee of the Madras Medical Mission, Chennai (Reg. No: ECR/140/Inst/TN/2013/RR-20).

**Informed Consent** The informed consent for each patient included in the study was obtained.

**Consent for Participation** The informed consent for each patient included in the study was obtained.

**Consent for Publication** All the authors are aware and have no conflicts for publication.

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